

4. When magnesium ribbon is added to excess hydrochloric acid, HCl, magnesium chloride and hydrogen gas are produced.



(a) Suggest **two** things that you would expect to **see** during this reaction. [2]

1.
2.

(b) Write the **symbol** equation for the reaction that takes place. [3]



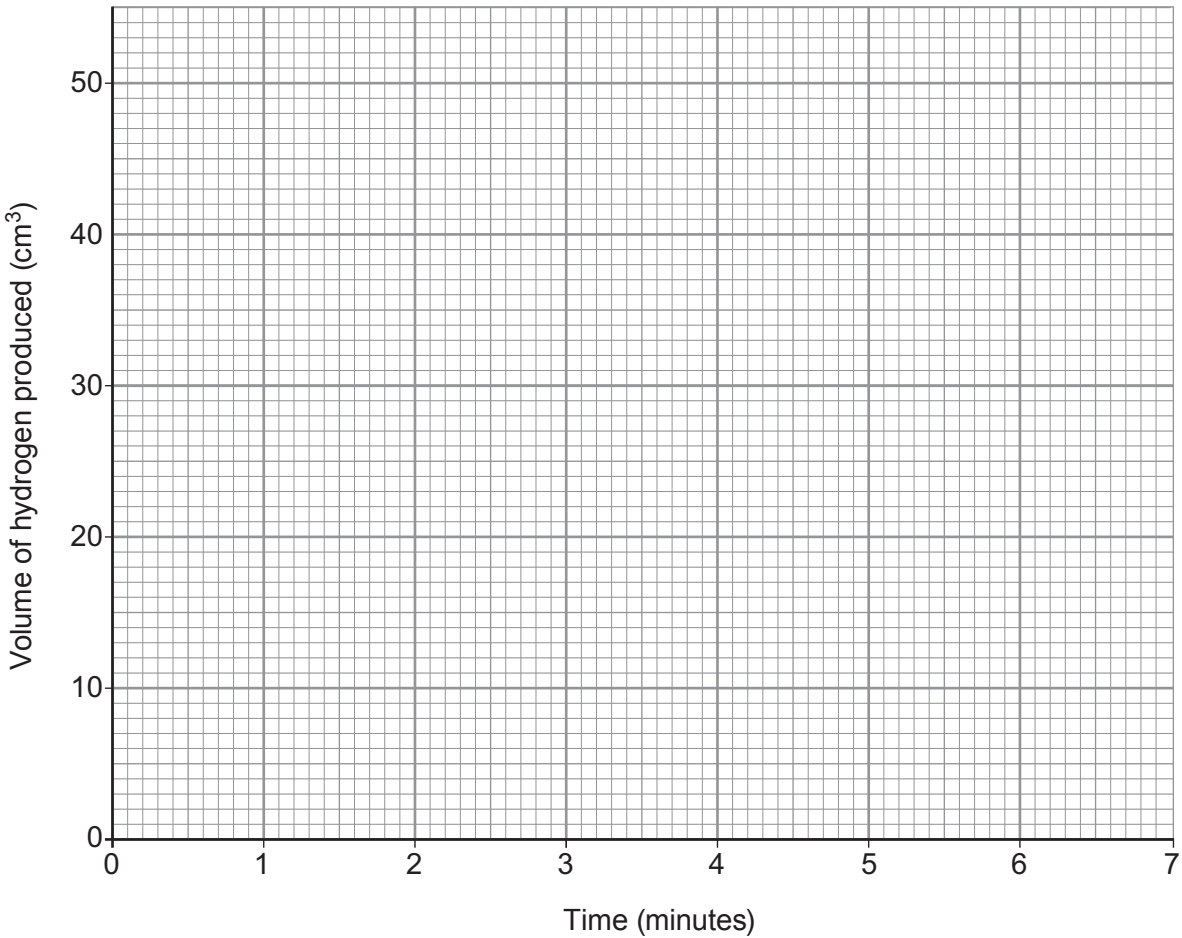
(c) Amelia recorded the volume of gas produced for 7 minutes for this reaction. She carried out the reaction at room temperature, 20 °C. The results she collected are shown in the table.

Time (minutes)	Volume of hydrogen produced (cm ³)
0	0
1	15
2	28
3	32
4	45
5	49
6	50
7	50

(i) Plot a graph of her results **on the grid opposite**. Draw a suitable line and label it **A**. [3]



Examiner
only



- (ii) Use the graph to find:
- I. the volume of gas produced in 90 seconds, [1]
 - II. the time taken for the reaction to end. [1]
- (iii) **On the same grid**, sketch the graph that would be obtained if Amelia repeated the same experiment at 40 °C. Label it **B**. [2]

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4. A student investigated changes that happened when pairs of colourless solutions were mixed. The table shows the observations made after mixing.

Pair of solutions	Appearance of the reactants	Appearance when mixed	Temperature change when mixed
A	two colourless solutions	no change	increase
B	two colourless solutions	bubbles form	increase
C	two colourless solutions	white precipitate forms	no change
D	two colourless solutions	no change	no change

- (a) Which pair of solutions do **not** react? Give a reason for your answer. [2]

.....

.....

.....

- (b) (i) The gas given off when pair **B** react is carbon dioxide. Describe the test for carbon dioxide gas. [1]

.....

.....

- (ii) One of the solutions in pair **B** is sodium carbonate. Give the formula of sodium carbonate. [1]

.....

- (c) (i) Which pair of solutions is silver nitrate and sodium chloride? [1]

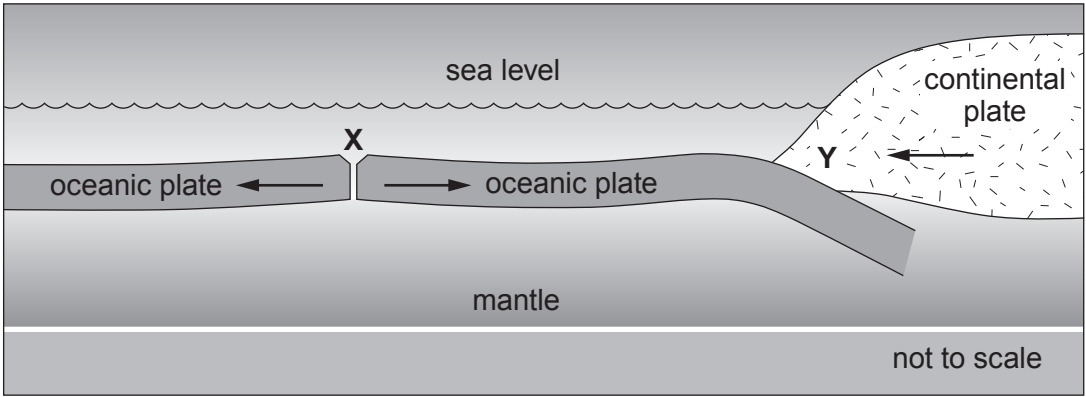
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- (ii) What colour would sodium chloride give in a flame test? [1]

.....



4. The Earth’s crust is broken up into tectonic plates.
- The diagram shows two plate boundaries **X** and **Y**.



- (a) Describe what happens as the plates move apart at plate boundary **X**. [2]

.....

.....

.....



(b) The two types of plate have different densities as shown in the table.

Type of plate	Density (g/cm ³)
continental	4.7
oceanic	5.0

At boundary **Y** the oceanic plate is pushed **underneath** the continental plate.

(i) State why the oceanic plate is pushed underneath the continental plate. [1]

.....

(ii) State what happens to the oceanic plate when it goes underneath the continental plate. [1]

.....

(iii) Underline the name of the type of plate boundary seen at **Y**. [1]

destructive constructive conservative

(c) Using satellite imaging, scientists discovered that one oceanic plate had moved 537 cm over 600 years. Calculate the mean rate of movement of the plate over this time. [2]

rate = $\frac{\text{distance}}{\text{time}}$

Rate = cm/year

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4. The table shows various stages in the development of the Earth’s atmosphere since its formation 4 500 million years ago.

Stage	Major events	Gases present in the atmosphere
1	volcanic eruptions	carbon dioxide, water vapour, methane, ammonia
2	oceans form	carbon dioxide, methane, ammonia
3	green plants evolve	carbon dioxide, nitrogen, oxygen
4	most carbon dioxide becomes locked in rock and fossil fuels	nitrogen, oxygen, water vapour

(a) Which **one** of these statements best describes how the oceans were formed?

Put a tick (✓) in the box next to the correct answer.

[1]

Water vapour evaporated to form clouds

☐

The Earth cooled so water vapour condensed

☐

Bacteria and algae turned the water vapour into liquid water

☐

There were no more volcanoes to produce water vapour

☐

(b) Explain why the appearance of green plants was an important stage in the development of the atmosphere. [2]

.....

.....

.....



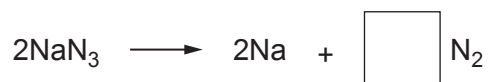
- (c) The atmosphere today contains nitrogen, oxygen and water vapour. [1]

Use the following information to identify another gas present.

- a Group 0 gas
 - the third most abundant in the atmosphere
 - used in light bulbs and as an inert atmosphere for welding
-

- (d) Nitrogen can be obtained by heating sodium azide, NaN_3 . Sodium is also produced in the reaction.

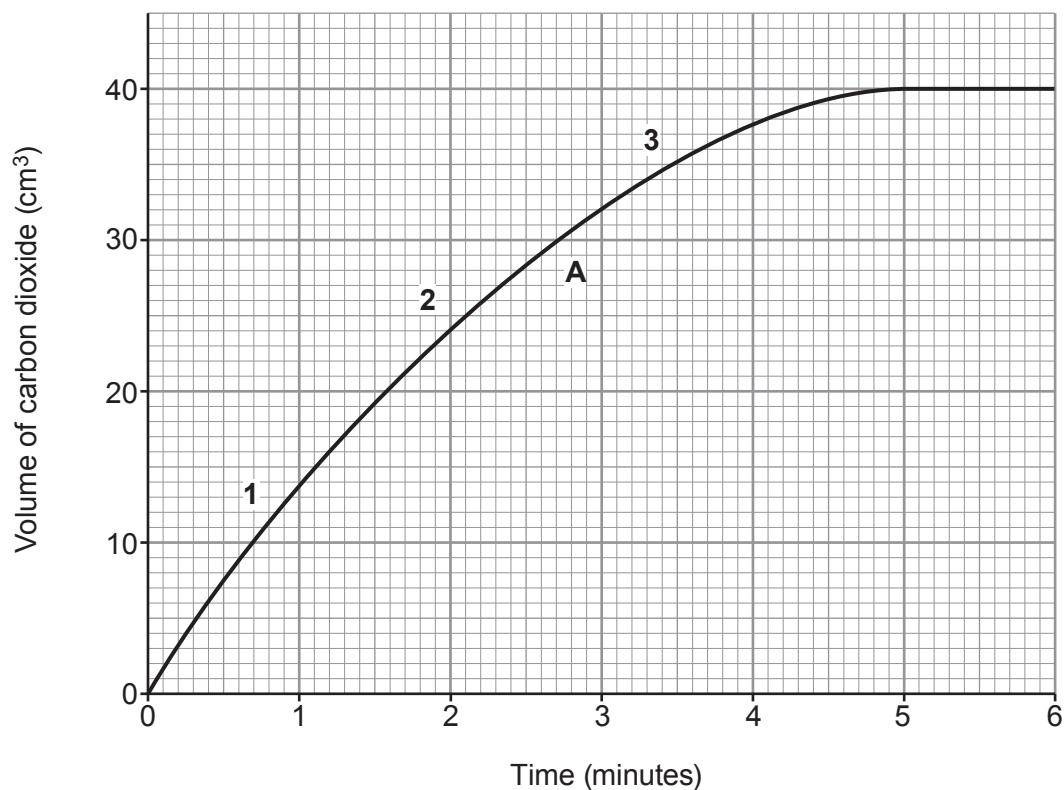
Complete the balancing of the equation for this reaction. [1]



4. Marble chips (calcium carbonate) react with hydrochloric acid to produce carbon dioxide gas as shown in the equation.



Graph **A** shows the volume of carbon dioxide produced over 6 minutes when the reaction was carried out at 30 °C.



- (a) (i) Find the volume of carbon dioxide given off after 2 minutes. [1]

..... cm³

- (ii) Find the time taken for the reaction to finish. [1]

..... minutes

- (iii) At which of the following points on graph **A** is the reaction at its fastest? [1]

Tick (✓) the correct box.

1

☐

2

☐

3

☐


- (iv) After $\frac{1}{2}$ minute, 7.5 cm^3 of carbon dioxide has been produced.

Use the equation to calculate the mean rate of reaction over this time in cm^3/s .

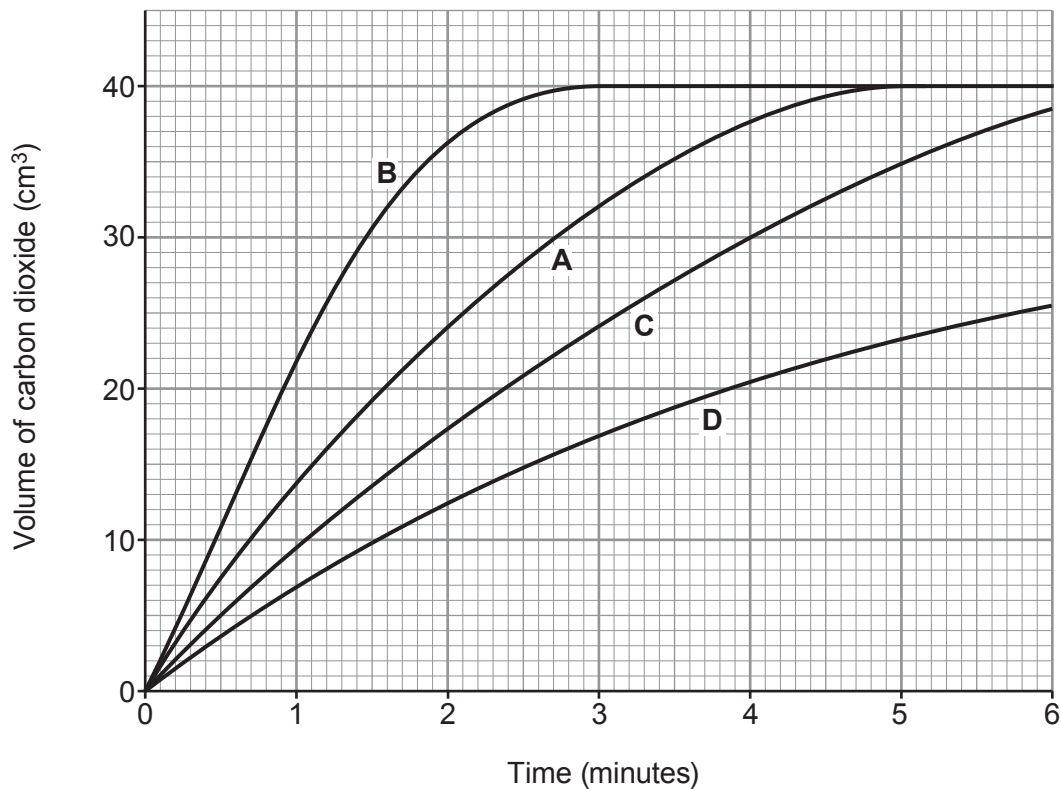
[2]

$$\text{mean rate} = \frac{\text{volume of carbon dioxide produced}}{\text{time (in seconds)}}$$

Mean rate = cm^3/s



- (b) (i) Graphs **B**, **C** and **D** on the grid below show the results for the experiment carried out at **different** temperatures. Graph **A** for the reaction carried out at 30 °C is also shown.



State which of graphs **B**, **C** and **D** shows the results for the experiment carried out at a temperature of 40 °C. [1]

Graph

- (ii) Using your knowledge of particle theory, underline the correct words in the brackets to complete the following sentence. [2]

At a higher temperature, the particles will have

(**more** / less / the same amount of) energy, so they will move faster and

there will be (an equal / a smaller / a greater) number of successful

collisions per second.

